

Voronoi-Based Energy-Aware User Partitioning and DRL Resource Allocation in Dual-UAV RIS-Assisted SWIPT Networks

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Abstract—Sixth-generation (6G) wireless networks are expected to support massive Internet of Things (IoT) connectivity, particularly for users located in remote or disaster-affected areas where terrestrial infrastructure is unavailable. Unmanned aerial vehicles (UAVs), owing to their high mobility and flexible deployment, are natural candidates for providing such on-demand coverage. When combined with reconfigurable intelligent surfaces (RISs), UAVs can further mitigate channel blockages and enhance spectral efficiency. However, the service lifetime of UAV-RIS systems remains fundamentally limited by on-board battery capacity and the power consumption of active RIS elements. To improve energy sustainability, we consider a dual-UAV RIS-assisted SWIPT network operating in a compact cell and develop a framework that couples energy-aware user partitioning with online resource control. First, we design a battery-aware weighted Voronoi scheme that adjusts service regions based on both user distribution and a normalized residual energy indicator for each UAV, thereby achieving proactive load and energy balancing. Building on the resulting trajectories and partitions, we then employ deep reinforcement learning (DRL) to jointly optimize the SWIPT time-switching factor, per-user transmit power, and RIS reflection configuration over a finite horizon of time slots. Simulation results demonstrate that the proposed dual-UAV architecture significantly improves energy-harvesting efficiency, average throughput, and worst-case user throughput compared with conventional single-UAV and non-weighted partition baselines.

Index Terms—Unmanned Aerial Vehicle (UAV), Reconfigurable Intelligent Surface (RIS), SWIPT, Weighted Voronoi Diagram, Deep Reinforcement Learning (DRL).

I. INTRODUCTION

The evolution toward sixth-generation (6G) wireless networks is driven by the imperative to connect a massive number of Internet of Things (IoT) devices, particularly in infrastructure-scarce environments. Unmanned aerial vehicles (UAVs) have emerged as agile aerial platforms capable of providing on-demand coverage owing to their high mobility and flexible deployment [1], [2]. To further mitigate line-of-sight (LoS) blockage, recent research has integrated reconfigurable intelligent surfaces (RIS) into UAV systems, where UAV-mounted RIS units can adaptively reshape the propagation environment [3], [4]. Despite these advantages, the operational endurance of UAV-RIS systems is constrained by finite battery capacity and the power consumption of RIS control circuitry. To extend the mission lifetime, simultaneous wireless information and power transfer (SWIPT) allows devices to harvest

energy from radio-frequency signals while maintaining data transmission [5]–[7]. For instance, Peng *et al.* [8], [9] proposed a resource-allocation-based harvest-transmit-store (RAHTS) model using the SD3 algorithm for a single UAV-RIS system. Subsequent studies have leveraged various deep reinforcement learning (DRL) frameworks, such as TD3 [10], [11] and PPO [12], to enhance energy harvesting (EH) efficiency.

Most existing works, however, focus on *single-UAV* scenarios [8]–[17]. When users are geographically dispersed, a single UAV faces a severe trade-off between distance and throughput. While deploying multiple agents can improve spatial coverage [18]–[20], it introduces the challenge of *load imbalance* among UAVs with heterogeneous battery states. Classical geometric partitioning, such as K-means, divides users solely by proximity [9], [21], ignoring the residual “mission energy” of each UAV. This can lead to premature battery depletion of overloaded nodes. Although load-aware partitioning via power diagrams has been explored in UAV-enabled MEC [22], its application in multi-UAV SWIPT networks to balance EH requirements and communication throughput remains largely unaddressed.

Motivated by this, we propose a collaborative *dual-UAV* RIS-assisted SWIPT framework. In the first stage (P1), we generate trajectories and service regions using a battery-aware weighted Voronoi strategy that proactively balances load based on user density and normalized residual energy. In the second stage (P2), a DRL agent adaptively controls the SWIPT time-switching (TS) ratio, transmit power, and RIS phase shifts. The objective is to maximize EH efficiency and sum-throughput while ensuring fairness under quality of service constraints.

The main contributions are summarized as follows:

- 1) **System Model:** We formulate a collaborative dual-UAV RIS-assisted SWIPT system to overcome the spatial coverage and resource bottlenecks inherent in conventional single-UAV architectures. Unlike single-UAV works [8]–[17], our model explicitly addresses the challenges of serving mobile users under a shared power budget. The primary objective of our system is to maximize the energy harvesting (EH) efficiency, while simultaneously enhancing the sum-throughput and the worst-case (minimum) user throughput as crucial secondary metrics.
- 2) **Offline Strategy:** We introduce an offline, battery-aware weighted Voronoi partitioning strategy. Unlike purely proximity-based K-means clustering [9], [21], this scheme dynamically adjusts the service regions

by jointly considering the spatial user density and the normalized residual energy of each UAV. Furthermore, by incorporating a rotary-wing UAV propulsion energy consumption model [23], our method achieves proactive energy-aware load balancing to prevent premature battery depletion.

- 3) **Online Algorithm:** We develop a DRL-based online controller to optimize dual-domain SWIPT variables within the fixed geometric partitions. Specifically, the DRL agent jointly optimizes the time-switching factor (τ), transmit power allocation (p), RIS phase shifts (θ), and RIS element switching states (w). By decoupling the large-scale spatial deployment (P1) from the small-scale signal-level resource allocation (P2), we maintain a tractable search space and achieve highly efficient online control.
- 4) **Simulation:** We conduct extensive simulations employing a rigorous two-stage evaluation process, where the training and testing trajectories are strictly separated. The results demonstrate that when confronted with unseen user mobility paths, our proposed scheme significantly outperforms traditional single-UAV baselines and non-weighted dual-UAV methods in terms of EH efficiency, average sum-throughput, and worst-case user throughput.

II. RELATED WORK

In this paper, we have reviewed several significant related works, and Table I provides a comparative analysis between our study and the referenced articles.

A. RIS-Assisted Signal Transmission

RIS technology has revolutionized wireless propagation by establishing virtual Line-of-Sight (LoS) links to overcome blockage. Shi *et al.* [4] highlighted the potential of RIS in cell-free massive MIMO systems to enhance spectral efficiency. In aerial scenarios, UAV-mounted RISs offer greater flexibility than terrestrial fixed RISs. For instance, Zargari *et al.* [24] optimized UAV trajectory and user scheduling to minimize energy consumption while ensuring quality of service (QoS). Recent advancements also explore Active RIS architectures; Aiswarya *et al.* [17] and Wang *et al.* [16] demonstrated that active elements can compensate for double-path loss attenuation, though at the cost of higher onboard power consumption, necessitating more efficient energy management strategies.

B. RIS-Assisted Energy Harvesting and SWIPT

To address the limited operational endurance of UAVs, SWIPT has been extensively studied as a key enabling technology [5]. **Single UAV-RIS Enabled SWIPT Systems:** Pioneering works by Peng *et al.* [8] introduced a dual-domain resource allocation scheme (RAHTS) that spatially splits RIS elements for simultaneous reflection and energy harvesting. This was extended in [9] to maximize energy harvesting efficiency (EHE) via a robust SD3 framework. Subsequent studies optimized various metrics: Zhang *et al.* [11] focused on secrecy EHE

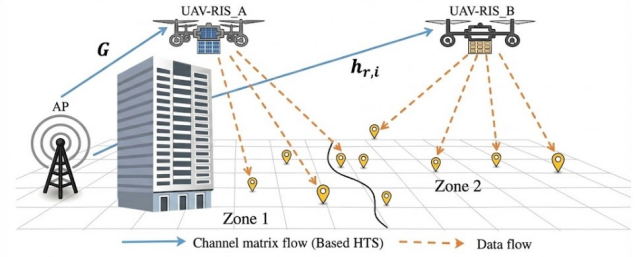


Fig. 1: Dual-UAV RIS-assisted SWIPT system architecture with zone-based service geometry.

using TD3, while Rathore *et al.* [15] utilized SD3 to handle high-dimensional state spaces. However, these single-UAV architectures [8]–[17] are fundamentally limited by the *near-far effect* in large-scale deployments, where service quality degrades for users distant from the UAV’s trajectory.

C. Deep Reinforcement Learning for RIS Systems

Given the non-convex nature of joint trajectory and resource optimization, Deep Reinforcement Learning (DRL) has emerged as a robust solution. Deep Deterministic Policy Gradient (DDPG) was early adopted by Peng *et al.* [8] to handle continuous action spaces in single-UAV systems. While more complex variants like TD3 [10], [11] and SD3 [9], [13], [15] have been proposed to mitigate Q-value overestimation, they often introduce higher computational overhead. In this paper, we leverage the computational efficiency of the DDPG framework and demonstrate that, when coupled with a well-designed battery-aware geometric partition, it can effectively manage the cooperative resource allocation in dual-UAV networks without the excessive complexity of twin-critic architectures.

D. Multi-UAV Cooperation and Trajectory Planning

To overcome the coverage constraints of single-UAV systems, cooperative architectures have gained significant interest. Recent works have explored Multi-Agent DRL (MADRL) for static RIS beamforming [25] and UAV-based sensing [26]. However, a critical gap exists in load-balanced trajectory design. Most existing multi-UAV studies rely on heuristic partitioning methods, such as K-means [14] or geometric centroids [15], which consider only spatial proximity. While Optimal Transport Theory [22] offers a theoretical basis for load balancing in MEC, it has not been fully exploited in the context of SWIPT-enabled UAV-RIS networks. Our work bridges this gap by integrating a Battery-Aware Weighted Voronoi partitioning strategy with a DDPG-based resource controller, ensuring that UAVs with lower energy levels are assigned lighter service loads to prolong the overall network lifetime.

III. SYSTEM MODEL

As illustrated in Fig. 1, we consider a dual-UAV RIS-assisted SWIPT network serving K single-antenna user terminals (UTs), indexed by $\mathcal{K} \triangleq \{1, \dots, K\}$. The access point

TABLE I: Comparison of Related Works and This Work

Ref.	Multi-RIS	Deployment	RIS Type	Multi-User	EH Domain	Optimization Scheme	Trajectory / Resource Allocation	Objective	Energy Aware
[8]	-	Static	Passive	-	Dual (Time+Space)	DDPG (RL)	-	Max Harvested Energy	-
[9]	-	Dynamic	Passive	✓	Dual (Time+Space)	SD3 (RL)	-	Max EH Efficiency	-
[12]	-	Dynamic	Passive	✓	Dual (Time+Space)	PPO (RL)	-	Max EH Efficiency	-
[10]	-	Dynamic	Passive	✓	Dual (Time+Space)	TD3 (RL)	NNS	Max EH Efficiency	-
[25]	✓	Static	Passive	✓	Time-only	MADRL	-	Max EH Efficiency	-
[13]	-	Dynamic	Passive	✓	Dual (Time+Space)	AE-SD3 (RL)	-	Max EH Efficiency	-
[11]	-	Dynamic	Passive	✓	Time-only	TD3 (RL)	-	Max Secrecy EH Efficiency	-
[14]	-	Dynamic	Passive	✓	Dual (Time+Space)	PER-SD3 (RL)	K-means / Fermat Point	Max EH Efficiency + Throughput	-
[15]	-	Dynamic	Passive	✓	Dual (Time+Space)	PPO (RL)	Geometric Centroid	Max EH Efficiency	-
[16]	-	Dynamic	Active	✓	Dual (Time+Space)	RL	-	Max Secrecy EE	✓
[17]	-	Dynamic	Active	✓	Dual (Time+Space)	Hybrid RL	-	Max EH Efficiency	-
This Work	✓	Dynamic	Passive	✓	Dual (Time+Space)	DDPG (RL)	Weighted Voronoi Partition	Max EH Efficiency + Fairness	✓

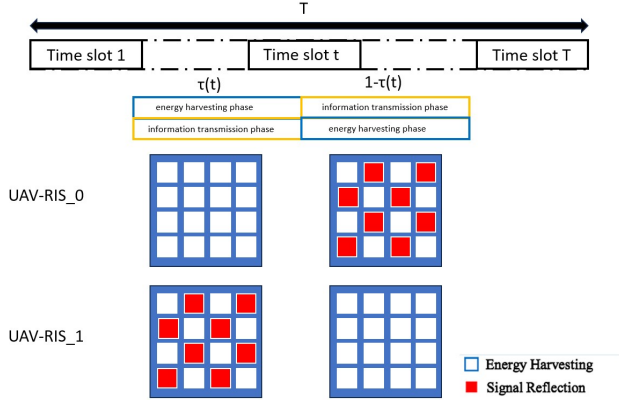


Fig. 2: Proposed complementary harvest–transmit–store (RAHTS) protocol [8], [9].

(AP) is equipped with Z antennas. Two UAVs, indexed by $u \in \mathcal{U} \triangleq \{1, 2\}$, are deployed at fixed altitudes H_u , and each carries a reconfigurable intelligent surface (RIS) consisting of L passive reflecting elements to intelligently reshape the wireless propagation environment [?], [27]. The system operates over a finite horizon of T discrete time slots indexed by $\mathcal{T} \triangleq \{1, \dots, T\}$. To prolong the operational endurance of the UAVs, the system adopts the RAHTS protocol [9] to perform simultaneous information reflection and energy harvesting in both the time and space domains, as depicted in Fig. 2.

A. AP Transmission and Air-to-Ground Channel Model

Following [27], the AP employs linear precoding and transmits

$$\mathbf{x}(t) = \sum_{k \in \mathcal{K}} \mathbf{v}_k(t) s_k(t), \quad (1)$$

where $\mathbf{v}_k(t) \in \mathbb{C}^{Z \times 1}$ is the beamforming vector for UT k , and $s_k(t) \sim \mathcal{CN}(0, 1)$ denotes the corresponding information symbol. The total AP transmit power is constrained as

$$\sum_{k \in \mathcal{K}} \|\mathbf{v}_k(t)\|^2 \leq P_{\max}. \quad (2)$$

The air-to-ground (AtG) [28], [29] channel between the AP and the ℓ -th RIS element mounted on UAV u is denoted by $\mathbf{g}_{u,\ell}(t) \in \mathbb{C}^{Z \times 1}$. The large-scale path loss is modeled as [28], [29]

$$PL_{u,\ell}(t) = \left(P_{u,\ell}^{\text{LoS}}(t) + (1 - P_{u,\ell}^{\text{LoS}}(t)) \kappa_{\text{NLoS}} \right) \times (\sqrt{x_{u,\ell}^2(t) + y_{u,\ell}^2(t) + H_u^2})^{-\alpha}, \quad (3)$$

where α is the path-loss exponent and κ_{NLoS} denotes the additional attenuation factor of NLoS links. The LoS probability is given by [30]

$$P_{u,\ell}^{\text{LoS}}(t) = \frac{1}{1 + A \exp(-B(\theta_{u,\ell}(t) - A))}, \quad (4)$$

where A and B are environment-dependent constants [31], and

$$\theta_{u,\ell}(t) = \frac{180}{\pi} \arcsin \left(\frac{H_u}{\sqrt{x_{u,\ell}^2(t) + y_{u,\ell}^2(t) + H_u^2}} \right) \quad (5)$$

denotes the elevation angle.

B. Resources Allocation Based Harvest–Transmit–Store Model

To enhance the endurance of the dual-UAV system, a complementary harvest–transmit–store (RAHTS) protocol is adopted. At each time slot t , the operation is divided into two phases controlled by a time-switching factor $\tau(t) \in [0, 1]$, as shown in Fig. 2. Specifically, one UAV performs information transmission while the other focuses on energy harvesting, and the roles are swapped between the two phases.

Let $\omega_{u,\ell}^k(t) \in \{0, 1\}$ denote the RIS element scheduling indicator, where $\omega_{u,\ell}^k(t) = 1$ indicates that the ℓ -th RIS element of UAV u reflects signals toward UT k . Each RIS element can serve at most one UT, and the complementary spatial energy-harvesting indicator is defined as

$$\bar{\omega}_{u,\ell}(t) = 1 - \sum_{k \in \mathcal{K}} \omega_{u,\ell}^k(t). \quad (6)$$

The harvested energy at UAV u during slot t is expressed as

$$\mathcal{E}_u(t) = \sum_{\ell=1}^L \eta \xi_u(t, \ell) \|\mathbf{g}_{u,\ell}^H(t) \mathbf{x}(t)\|^2, \quad (7)$$

where $\eta \in (0, 1)$ is the RF-to-DC conversion efficiency, and $\xi_u(t, \ell)$ represents the effective time–space harvesting coefficient induced by the complementary RAHTS protocol.

Following [8], [9], the energy harvesting efficiency of UAV u is defined as

$$\mathcal{E}_u^{\text{eff}}(t) = \frac{\mathcal{E}_u(t)}{H_u(t)}, \quad (8)$$

where $H_u(t) = \sum_{\ell=1}^L \|\mathbf{g}_{u,\ell}^H(t) \mathbf{x}(t)\|^2$ denotes the total impinging RF energy.

C. AP-RIS-UT Channel Model and Achievable Throughput

During the information transmission phase, passive reflective beamforming is performed. Let $\mathbf{h}_{u,k}(t) \in \mathbb{C}^{L \times 1}$ and $\mathbf{G}_u(t) \in \mathbb{C}^{L \times Z}$ denote the channels from RIS- u to UT k and from the AP to RIS- u , respectively. Following [9], [27], the reflection coefficient matrix of the UAV-mounted RIS is defined as

$$\Phi = \text{diag}(\rho_1 e^{j\theta_1}, \rho_2 e^{j\theta_2}, \dots, \rho_L e^{j\theta_L}) \in \mathbb{C}^{L \times L}, \quad (9)$$

where $j = \sqrt{-1}$ denotes the imaginary unit, $\theta_\ell \in (0, 2\pi)$ represents the phase shift of the ℓ -th reflecting element, and $\rho_\ell \in (0, 1]$ denotes the corresponding amplitude reflection coefficient.

Incorporating RIS element scheduling, the effective cascaded channel from the AP to UT k via its serving UAV $u_k(t)$ is given by

$$\hat{\mathbf{h}}_k^H(t) = \mathbf{h}_{u_k,k}^H(t) \Phi_{u_k}(t) \mathbf{G}_{u_k}(t). \quad (10)$$

The RIS-UT channel $\mathbf{h}_{u,k}(t)$ follows a Rician fading model with factor K_{Rician} :

$$\mathbf{h}_{u,k}(t) = \sqrt{\frac{K_{\text{Rician}}}{1 + K_{\text{Rician}}}} \mathbf{h}_{u,k}^{\text{LoS}}(t) + \sqrt{\frac{1}{1 + K_{\text{Rician}}}} \mathbf{h}_{u,k}^{\text{NLoS}}(t). \quad (11)$$

The received signal at UT k is

$$y_k(t) = \hat{\mathbf{h}}_k^H(t) \mathbf{v}_k(t) s_k(t) + n_k(t), \quad (12)$$

where $n_k(t) \sim \mathcal{CN}(0, \sigma_k^2)$ denotes additive white Gaussian noise. As in [8], [9], [32], it is assumed that each UT can perfectly cancel interference from signals reflected toward other UTs. Accordingly, the received signal-to-noise ratio (SNR) is given by

$$\text{SNR}_k(t) = \frac{|\hat{\mathbf{h}}_k^H(t) \mathbf{v}_k(t)|^2}{\sigma_k^2}. \quad (13)$$

Due to the complementary scheduling, the achievable throughput of UT k is

$$\Gamma_k(t) = \begin{cases} (1 - \tau(t)) B \log_2(1 + \text{SNR}_k(t)), & u_k(t) = 1, \\ \tau(t) B \log_2(1 + \text{SNR}_k(t)), & u_k(t) = 2, \end{cases} \quad (14)$$

and the quality-of-service (QoS) constraint requires $\Gamma_k(t) \geq \Gamma_{\min}$ for all $k \in \mathcal{K}$.

IV. PROBLEM FORMULATION (ONLINE STAGE P2)

Following the offline trajectory and user partition design in Stage P1, The online stage aims to maximize the total energy harvesting efficiency of the dual-UAV RIS-assisted SWIPT network over a finite horizon of T time slots, while strictly satisfying the minimum throughput requirements of all UTs.

The AP transmits a common multiuser signal subject to a total power budget $P_{\max}$, which illuminates both UAV-mounted RISs simultaneously through independent AP-RIS links. Given the fixed partition indicators $\{a_{u,k}(t)\}$ obtained in Stage P1, The online resource allocation problem is formulated as follows:

$$\begin{aligned} \text{(P2):} \quad & \max_{\tau(t), \mathbf{P}(t), \omega(t), \theta(t)} \sum_{t=1}^T \sum_{u \in \mathcal{U}} \mathcal{E}_u^{\text{eff}}(t) \\ \text{s.t.} \quad & \text{C1: } \Gamma_k(t) \geq \Gamma_{\min}, \quad \forall k \in \mathcal{K}, \forall t \in \mathcal{T}, \\ & \text{C2: } 0 \leq \tau(t) \leq 1, \quad \forall t \in \mathcal{T}, \\ & \text{C3: } \sum_{k \in \mathcal{K}} p_k(t) \leq P_{\max}, \quad \forall t \in \mathcal{T}, \\ & \text{C4: } 0 \leq p_k(t) \leq P_{\max}^{(k)}, \quad \forall k \in \mathcal{K}, \forall t \in \mathcal{T}, \\ & \text{C5: } \omega_{u,\ell}^k(t) \in \{0, 1\}, \quad \forall u, \ell, k, t, \\ & \text{C6: } \sum_{k \in \mathcal{K}} \omega_{u,\ell}^k(t) \leq 1, \quad \forall u, \ell, t, \\ & \text{C7: } \omega_{u,\ell}^k(t) \leq a_{u,k}(t), \quad \forall u, \ell, k, t, \\ & \text{C8: } \theta_{u,\ell}(t) \in [0, 2\pi), \quad \forall u, \ell, t, \\ & \text{C9: } |e^{j\theta_{u,\ell}(t)}| = 1, \quad \forall u, \ell, t. \end{aligned} \quad (15)$$

The decision variables are defined as follows:

- $\tau(t)$ denotes the time-switching factor at time slot t .
- $\mathbf{P}(t) = \{p_1(t), \dots, p_K(t)\}$ is the transmit power allocation vector for all UTs.
- $\omega(t) = \{\omega_{u,\ell}^k(t)\}$ denotes the RIS element scheduling variables.
- $\theta(t) = \{\theta_{u,\ell}(t)\}$ collects the RIS phase-shift variables for both UAV-mounted RISs.

Constraints C1–C2 ensure the QoS requirement and the valid range of the time-switching factor. Constraints C3–C4 impose the total and per-user AP transmit power limits. Constraints C5–C7 enforce the binary nature of RIS element scheduling and the consistency with the offline Voronoi-based user partition. Constraints C8–C9 correspond to the continuous phase-shift and unit-modulus constraints of passive RIS elements.

The optimization problem (P2) is formulated as a highly non-convex Mixed-Integer Non-Linear Programming (MINLP) problem. Its intractability primarily arises from the binary RIS scheduling variables (C5), the unit-modulus phase-shift constraints (C9), and the intricate coupling among continuous variables ($\tau(t), \mathbf{P}(t), \theta(t)$). Consequently, standard convex solvers are ineffective. Although traditional approaches such as alternating optimization (AO) or successive convex approximation (SCA) can address this via mathematical relaxations, their prohibitive computational complexity and slow convergence render them impractical for real-time decision-making in dynamic UAV networks. To circumvent these limitations and efficiently tackle the non-convexity, we propose a robust Deep Reinforcement Learning (DRL)-based framework in the following section.

V. OFFLINE STAGE (P1): TRAJECTORY AND PARTITION DESIGN

This study considers human mobility in dynamic service scenarios. Therefore, the UAV-RISs must periodically re-deploy to provide seamless coverage for mobile UTs. Following [9], each UAV-RIS is assumed to operate at a fixed altitude and move horizontally in the Cartesian coordinate system. In

general, reducing the total path loss is equivalent to shortening the Euclidean distances between UAVs, RISs, and UTs.

A. Problem Definition of Offline Stage P1

At each time slot t , the offline stage P1 focuses on determining the UAVs' horizontal positions and the corresponding user partitions, jointly considering service demand and energy sustainability, inspired by optimal transport theory for load balancing [22]. Specifically, P1 can be interpreted as a per-slot geometry-aware design problem that maximizes a weighted utility function:

$$\begin{aligned}
 \text{(P1): } & \max_{\{\mathbf{Q}_u(t), \mathcal{S}_u(t)\}} \sum_{u=1}^2 \left[(1 - \lambda) D_u(t) + \lambda B_u(t) \right] \\
 \text{s.t. } & \mathcal{S}_1(t) \cup \mathcal{S}_2(t) = \mathcal{K}, \\
 & \mathcal{S}_1(t) \cap \mathcal{S}_2(t) = \emptyset, \\
 & k \in \mathcal{S}_u(t) \Rightarrow \|\mathbf{C}_k(t) - \mathbf{Q}_u(t)\|^2 - r_u(t) \\
 & \leq \|\mathbf{C}_k(t) - \mathbf{Q}_v(t)\|^2 - r_v(t), \quad \forall v \neq u, \\
 & B_u(t) \geq B_{\min}, \quad \forall u \in \mathcal{U}, \\
 & r_{\min} \leq r_u(t) \leq r_{\max}, \quad \forall u \in \mathcal{U}.
 \end{aligned} \tag{16}$$

In (16), $D_u(t)$ denotes the normalized service load (density) of UAV u , and $B_u(t)$ represents the normalized residual battery level. The balancing factor $\lambda \in [0, 1]$ controls the tradeoff between spatial demand tracking and energy sustainability. The constraints ensure partition completeness and disjointness, enforce power diagram consistency, and guarantee a battery safety margin and bounded weight, respectively. It is worth noting that P1 is formulated on a per-slot basis and serves as a geometry-aware design principle rather than a long-term trajectory optimization.

B. Baseline: Distance-Aware Deployment (Single-UAV)

As a benchmark, we consider a conventional distance-aware deployment strategy for the single-UAV case [9], [12]. Specifically, the UAV position is determined by minimizing the sum of squared Euclidean distances to all associated users, which is equivalent to the K-means clustering criterion. The optimal horizontal position $\hat{\mathbf{Q}}(t)$ is obtained by solving:

$$\min_{\hat{\mathbf{Q}}(t)} \sum_{k \in \mathcal{K}} \|\mathbf{C}_k(t) - \hat{\mathbf{Q}}(t)\|^2, \tag{17}$$

where $\mathbf{C}_k(t)$ denotes the horizontal location of UT k . Although effective in reducing path loss, this method ignores the UAV battery state and may lead to premature energy depletion.

C. Proposed: Battery-Aware Weighted Voronoi Deployment (Dual-UAV)

In the collaborative dual-UAV scenario, pure geometric proximity is insufficient, as it may lead to severe load imbalance and energy exhaustion when UAVs have heterogeneous battery levels. To address this issue, we propose a *battery-aware weighted Voronoi (power diagram)* deployment strategy.

Remark (Link to Objective P1): Although the optimization problem (P1) is formulated as a system-level utility maximization, solving it directly is mathematically intractable due to the discrete, non-differentiable indicator function in $D_u(t)$ and the highly non-convex coupling between the UAV trajectories $\mathbf{Q}_u(t)$ and the user partitions $\mathcal{S}_u(t)$. Therefore, rather than seeking a highly complex global optimum, we propose this low-complexity, geometry-driven heuristic. It conceptually aligns with the objective of (P1) through an alternating perspective:

- **Partitioning via Power Diagram:** By introducing a density- and battery-dependent bias $r_u(t)$, the weighted Voronoi partition inherently expands the coverage of UAVs with higher residual energy or surrounding denser user clusters, aligning with the maximization of $B_u(t)$ and $D_u(t)$.
- **Trajectory via Centroid Update:** By moving the UAV toward the geometric centroid of its assigned users, the intra-cell distances are minimized. This intuitively pushes the UAV to cover more users within its service radius R_s , thereby increasing the density indicator $D_u(t)$.

The service region is partitioned into cells $\mathcal{S}_u(t)$ based on the weighted (power) distance:

$$\begin{aligned}
 k \in \mathcal{S}_u(t) & \iff \|\mathbf{C}_k(t) - \mathbf{Q}_u(t)\|^2 - r_u(t) \\
 & \leq \|\mathbf{C}_k(t) - \mathbf{Q}_v(t)\|^2 - r_v(t), \quad \forall v \neq u,
 \end{aligned} \tag{18}$$

where $r_u(t)$ denotes the weight (power bias) associated with UAV u . A larger $r_u(t)$ effectively enlarges the service region of UAV u .

1) *Weight Design (Density + Battery):* To balance service demand and energy sustainability, we define a density-based demand indicator and a residual-battery indicator, respectively. The service-demand (density) indicator is defined as

$$D_u(t) = \frac{1}{K} \sum_{k=1}^K \mathbf{1} \{ \|\mathbf{C}_k(t) - \mathbf{Q}_u(t)\| \leq R_s \} \in [0, 1], \tag{19}$$

where R_s is a predefined service radius. The normalized residual battery level is denoted by $B_u(t) \in [B_{\min}, B_{\max}]$, where $B_{\max} = 0.95$ and $B_{\min} = 0.2$.

Then, the bounded weight is designed as

$$w_u(t) = (1 - \lambda) D_u(t) + \lambda B_u(t), \quad 0 \leq w_u(t) \leq 1, \tag{20}$$

and the power-diagram bias is

$$r_u(t) = \rho w_u(t), \quad 0 \leq r_u(t) \leq \rho, \tag{21}$$

where ρ is a scaling factor (units of m^2) controlling the influence of the weight in (18).

Remark (Dynamic Tradeoff Mechanism): It is worth noting that the term $(1 - \lambda) D_u(t)$ encourages a “greedy” expansion toward dense user clusters to maximize service coverage. While this might temporarily increase the burden on a specific UAV, it inherently accelerates its energy consumption. As the battery $B_u(t)$ drops, the positive weight contribution from $D_u(t)$ is naturally counteracted by the decreasing $B_u(t)$. This negative feedback loop acts as an autonomous brake,

preventing any single UAV from fatal energy depletion and eventually forcing it to shrink its service cell, autonomously offloading users to the collaborative UAV.

2) *Trajectory Update with Mobility Constraint*: Given the weighted Voronoi partition, each UAV moves toward the geometric centroid of its assigned cell:

$$\hat{\mathbf{Q}}_u(t) = \frac{1}{|\mathcal{S}_u(t)|} \sum_{k \in \mathcal{S}_u(t)} \mathbf{C}_k(t). \quad (22)$$

To enforce the per-slot mobility limit $d_{\max}$ and avoid unrealistic “teleportation”, the UAV moves toward its assigned centroid with a clipped displacement $\Delta \mathbf{Q}_u(t) = \hat{\mathbf{Q}}_u(t) - \mathbf{Q}_u(t)$:

$$\mathbf{Q}_u(t+1) = \mathbf{Q}_u(t) + \Delta \mathbf{Q}_u(t). \quad (23)$$

3) *Per-Slot Mobility and Rotary-Wing Energy Consumption*: The horizontal speed of UAV u at slot t is

$$v_u(t) = \frac{\|\mathbf{Q}_u(t+1) - \mathbf{Q}_u(t)\|}{\Delta T}, \quad (24)$$

where ΔT is the slot duration. The propulsion power consumption is modeled as [23]:

$$P_{\text{prop}}(v) = P_0 \left(1 + \frac{3v^2}{U_{\text{tip}}^2} \right) + P_i \left(\sqrt{1 + \frac{v^4}{4v_0^4}} - \frac{v^2}{2v_0^2} \right)^{\frac{1}{2}} + \frac{1}{2} \rho s A_{\text{rot}} v^3, \quad (25)$$

where ρ is the air density and A_{rot} is the rotor disc area. The corresponding propulsion energy during slot t is

$$E_u^{\text{prop}}(t) = P_{\text{prop}}(v_u(t)) \Delta T. \quad (26)$$

To provide a clear physical insight into the objective of (P1), the rationale behind the formulation is summarized as follows:

- **Decision Variables and Goal**: The problem jointly optimizes the UAV positions $\mathbf{Q}_u(t)$ and user partitions $\mathcal{S}_u(t)$ to find the best tradeoff between spatial demand tracking and energy sustainability.
- **High Density** ($D_u(t)$): Encourages the UAV to aggressively expand its service region toward densely populated areas to maximize coverage.
- **High Battery** ($B_u(t)$): Allows the system to confidently assign a larger service area to a UAV with abundant energy.
- **Low Battery** ($B_u(t)$): Acts as an autonomous brake; it shrinks the service partition of a UAV nearing its energy lower bound, inherently forcing it to offload users and preventing premature depletion.
- **Maximizing the Weighted Sum**: Achieves a dynamic equilibrium where UAVs greedily track user mobility but are continuously regularized by their own energy consumption.

Algorithm 1: Battery-Aware Weighted Voronoi Deployment for Offline Stage (P1)

Input : User trajectories $\{\mathbf{C}_k(t)\}_{t=1}^T$; initial UAV states $\mathbf{Q}_u(1), B_u(1)$; parameters $\lambda, \rho, R_s, d_{\max}, B_{\min}$.

Output: Offline partition indicators $\{\mathcal{S}_u(t)\}_{t=1}^T$ and UAV trajectories $\{\mathbf{Q}_u(t)\}_{t=1}^{T+1}$.

```

1 for  $t = 1$  to  $T$  do
2   foreach UAV  $u \in \{1, 2\}$  do
3     Calculate user density  $D_u(t)$  within coverage radius  $R_s$ ;
4     Compute power bias:
        $r_u(t) = \rho[(1 - \lambda)D_u(t) + \lambda B_u(t)]$ ;
5   end
6   Initialize partitions  $\mathcal{S}_1(t) = \emptyset$  and  $\mathcal{S}_2(t) = \emptyset$ ;
7   foreach user  $k \in \mathcal{K}$  do
8     Find optimal UAV  $u^*$  by minimizing the weighted distance:  $u^* = \arg \min_{v \in \{1, 2\}} (\|\mathbf{C}_k(t) - \mathbf{Q}_v(t)\|^2 - r_v(t))$ ;
9     Assign user  $k$ :  $\mathcal{S}_{u^*}(t) = \mathcal{S}_{u^*}(t) \cup \{k\}$ ;
10  end
11  foreach UAV  $u \in \{1, 2\}$  do
12    Compute target centroid  $\hat{\mathbf{Q}}_u(t)$  based on assigned partition  $\mathcal{S}_u(t)$ ;
13    Update position  $\mathbf{Q}_u(t+1)$  moving towards  $\hat{\mathbf{Q}}_u(t)$  subject to max flying distance  $d_{\max}$ ;
14    Update residual battery level  $B_u(t+1)$  based on energy consumption limit  $B_{\min}$ ;
15  end
16 end

```

VI. DEEP REINFORCEMENT LEARNING ALGORITHM-BASED FRAMEWORK

To optimally allocate communication and energy-harvesting resources under dynamic channel and user mobility, we adopt a deep reinforcement learning (DRL) framework to solve the online resource allocation problem (P2). Since the formulated problem is highly non-convex and involves coupled continuous and discrete variables, conventional optimization methods are computationally intractable for real-time implementation. Therefore, we employ the *Deep Deterministic Policy Gradient (DDPG)* algorithm as the core learning engine.

A. Generalized Deep Reinforcement Learning

Reinforcement learning is derived from the Markov decision process (MDP) [33] interaction between an intelligent agent and the external environment. The formulated MDP can be expressed as follows:

$$\mathcal{G} := \langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle, \quad (27)$$

where $\mathcal{S}$ and $\mathcal{A}$ represent the finite sets of states and actions, respectively; $\mathcal{P} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$ is the state transition probability; $\mathcal{R} : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ denotes the reward function; and $\gamma \in [0, 1]$ is the discount factor determining the importance of future rewards.

In the DDPG framework [34], which is an actor-critic method designed for continuous control, the agent maintains two primary networks:

- **Actor Network** $\mu(s|\theta^\mu)$: Maps the current state s_t to a specific deterministic action a_t .
- **Critic Network** $Q(s, a|\theta^Q)$: Estimates the action-value function (Q-value) to evaluate the actor's policy.

To ensure stability, DDPG also employs target networks μ' and Q' with parameters $\theta^{\mu'}$ and $\theta^{Q'}$, which slowly track the main networks via soft updates.

At each time step t , the agent executes an action $a_t = \mu(s_t|\theta^\mu) + \mathcal{N}_t$, where $\mathcal{N}_t$ is exploration noise. The tuple (s_t, a_t, r_t, s_{t+1}) is stored in a replay buffer $\mathcal{D}$. The critic is updated by minimizing the Bellman loss:

$$L(\theta^Q) = \frac{1}{N_b} \sum_{i=1}^{N_b} (y_i - Q(s_i, a_i|\theta^Q))^2, \quad (28)$$

where $y_i = r_i + \gamma Q'(s_{i+1}, \mu'(s_{i+1}|\theta^{\mu'})|\theta^{Q'})$ is the target value. The actor is updated using the deterministic policy gradient:

$$\nabla_{\theta^\mu} J \approx \frac{1}{N_b} \sum_{i=1}^{N_b} \nabla_a Q(s, a|\theta^Q)|_{s=s_i, a=\mu(s_i)} \nabla_{\theta^\mu} \mu(s|\theta^\mu)|_{s_i}. \quad (29)$$

B. Proposed DDPG-Based Resource Allocation Scheme

In our dual-UAV RIS-assisted SWIPT system, the DRL agent acts as the central controller. The interaction procedure is defined as follows:

1) *State Space (Observation)*: At each time slot t , the DRL agent observes the global system state s_t , which captures the current wireless environment, network geometry, and UAV battery status. The observation is defined as:

$$s_t = \left\{ \mathbf{G}_u(t), \mathbf{h}_{u,k}(t), d_{u,k}(t), \mathbf{Q}_u(t), \mathbf{C}_k(t), S_u(t), B_u(t) \right\}_{u \in \mathcal{U}, k \in \mathcal{K}}, \quad (30)$$

where:

- $\mathbf{G}_u(t)$ denotes the baseband-equivalent AP-to-UAV-RIS channel;
- $\mathbf{h}_{u,k}(t)$ is the UAV-RIS-to-UT k channel;
- $d_{u,k}(t)$ is the distance between UAV u and UT k ;
- $\mathbf{Q}_u(t)$ and $\mathbf{C}_k(t)$ are the horizontal coordinates of UAV u and UT k , respectively;
- $S_u(t)$ is the number of UTs assigned to UAV u by the P1 partition;
- $B_u(t)$ denotes the residual battery level of UAV u .

This state design explicitly incorporates the geometry-aware partitioning results from P1, enabling the DRL agent to perform resource allocation under fixed-per-slot UAV deployments.

2) *Action Space*: At each time slot t , the DRL agent outputs a continuous action vector a_t to control the dual-domain SWIPT operation. The action space is defined as:

$$a_t = \left\{ \tau(t), \{p_k(t)\}_{k \in \mathcal{K}}, \{\theta_{u,\ell}(t)\}_{u,\ell}, \{\tilde{\omega}_{u,\ell}^k(t)\}_{u,\ell,k} \right\}, \quad (31)$$

where:

- $\tau(t) \in [0, 1]$ is the time-switching factor between EH and IT phases;
- $p_k(t) \in [0, P_{\max}]$ denotes the transmit power allocated to UT k ;
- $\theta_{u,\ell}(t) \in [0, 2\pi]$ is the continuous phase shift of RIS element ℓ on UAV u ;
- $\tilde{\omega}_{u,\ell}^k(t) \in \mathbb{R}$ is the continuous relaxation of the binary RIS scheduling variable.

Action Mapping Strategy: Since RIS element scheduling is inherently discrete, the relaxed variable $\tilde{\omega}_{u,\ell}^k(t)$ is mapped to a binary decision $\omega_{u,\ell}^k(t) \in \{0, 1\}$ via a post-processing operation. Moreover, the mapping strictly enforces the P1 partition constraint, such that $\omega_{u,\ell}^k(t) = 0$ if UT k is not assigned to UAV u in the offline stage.

3) *Reward Function*: The positive reward represents the objective of the proposed framework, which is to maximize the overall energy harvesting efficiency of the UAV-RIS system. At each time step t , the instantaneous reward is positively correlated with the energy harvesting efficiency $\varepsilon(t)$. The proposed framework must also account for the users' minimum capacity requirements. Hence, the reward r_t can be described as follows:

$$r_t = \varepsilon(t) \times \rho(t), \quad (32)$$

where $\rho(t)$ acts as a hard QoS penalty factor indicating whether all UTs satisfy the required throughput $\Gamma_{\min}$, and is defined by:

$$\rho(t) = \prod_{k \in \mathcal{K}} \rho_k(t), \quad (33)$$

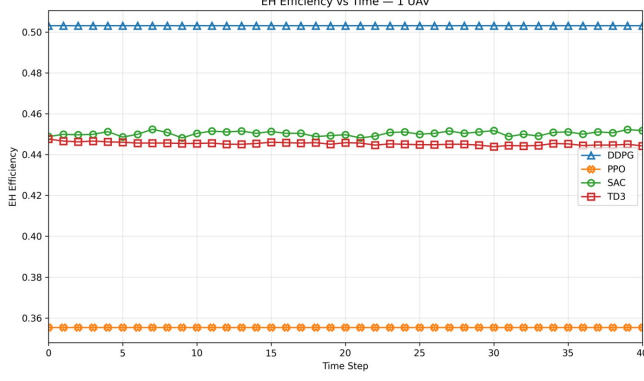
where $\rho_k(t)$ is given by:

$$\rho_k(t) = \begin{cases} 0, & \Gamma_k(t) < \Gamma_{\min}, \\ 1, & \Gamma_k(t) \geq \Gamma_{\min}. \end{cases} \quad (34)$$

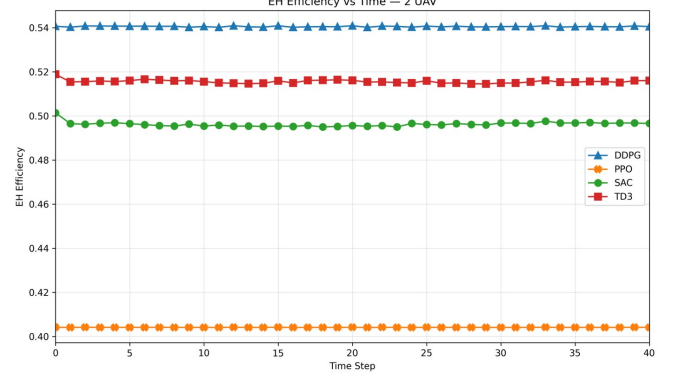
This reward structure forces the agent to jointly optimize energy harvesting performance and communication reliability.

C. Summary of the Learning Process

The entire training process is summarized in Algorithm 2. The agent iteratively interacts with the dual-UAV environment. In each step, it observes the composite state s_t , generates continuous actions via the Actor network, maps them to feasible discrete scheduling decisions, and receives the reward r_t . The Actor and Critic networks are updated via gradient descent to maximize the long-term expected reward.

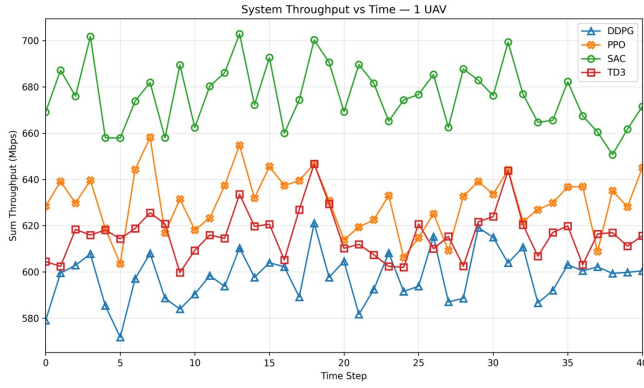


(a) Single-UAV System

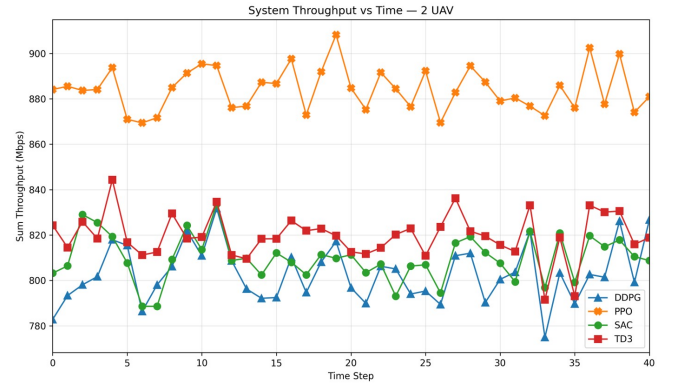


(b) Dual-UAV System

Fig. 3: Energy harvesting efficiency per testing step for (a) single-UAV and (b) dual-UAV systems under different DRL algorithms.

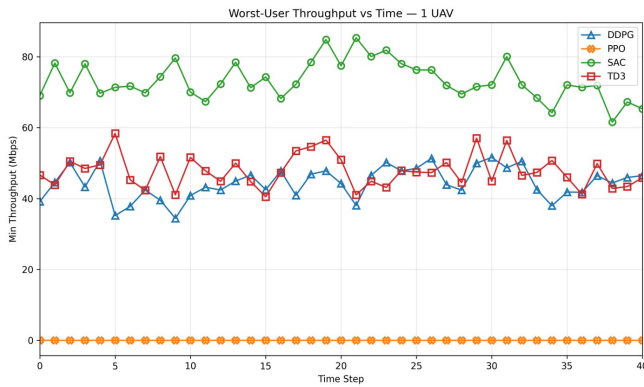


(a) Single-UAV System

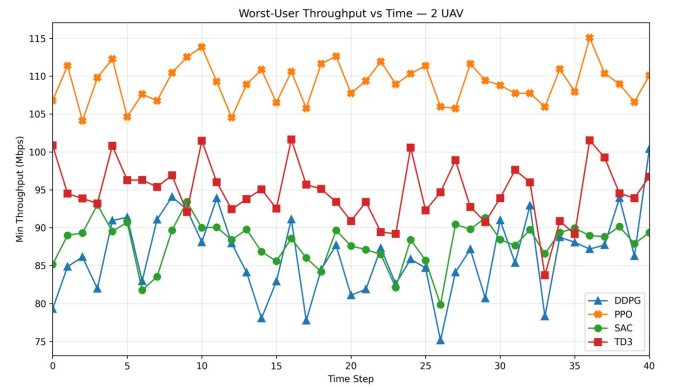


(b) Dual-UAV System

Fig. 4: System sum throughput per testing step for (a) single-UAV and (b) dual-UAV systems.



(a) Single-UAV System



(b) Dual-UAV System

Fig. 5: Worst-user throughput versus testing step for (a) single-UAV and (b) dual-UAV systems under different DRL algorithms.

VII. SIMULATION RESULTS

In this section, the performance of the proposed DRL-based resource allocation framework is evaluated under dynamic user mobility scenarios. To rigorously verify the generalization and robustness of the proposed architecture, the simulation

procedure is systematically executed through the following three-step experimental flow:

- **Step 1: Offline Trajectory Generation (P1).** UAV trajectories and user partitions are first determined. The density-aware (K-means) method is used for the single-

Algorithm 2: DDPG-Based Online Resource Allocation for Problem (P2)

Input: Offline partition indicators $\{a_{u,k}(t)\}$ and UAV positions $\{\mathbf{Q}_u(t)\}$ from P1; channels $\{\mathbf{G}_u(t), \mathbf{h}_{u,k}(t)\}$; hyperparameters: replay buffer size N_D , minibatch size N_b , discount factor γ , soft update factor ψ .

Output: Learned policy $\mu(\cdot)$ for $a_t = \{\tau(t), \mathbf{P}(t), \Theta(t), \omega(t)\}$.

- 1 Initialize actor network $\mu(s|\theta^\mu)$ and critic network $Q(s, a|\theta^Q)$;
- 2 Initialize target networks μ' and Q' with $\theta^{\mu'} \leftarrow \theta^\mu$, $\theta^{Q'} \leftarrow \theta^Q$;
- 3 Initialize replay buffer $\mathcal{D}$;
- 4 **for** $episode = 1$ to N_{ep} **do**
- 5 Reset environment and obtain initial state s_1 ;
- 6 **for** $t = 1$ to T **do**
- 7 $\tilde{a}_t = \mu(s_t|\theta^\mu) + \mathcal{N}_t$;
- 8 Decode $\tilde{a}_t \rightarrow \{\tau(t), p_k(t), \theta_{u,\ell}(t), \tilde{\omega}_{u,\ell}^k(t)\}$;
- 9 Apply partition mask: $\tilde{\omega}_{u,\ell}^k(t) \leftarrow 0$ if $a_{u,k}(t) = 0$;
- 10 **foreach** (u, ℓ) **do**
- 11 **if** $\max_k \tilde{\omega}_{u,\ell}^k(t) > 0$ **then**
- 12 $k^* = \arg \max_k \tilde{\omega}_{u,\ell}^k(t)$;
- 13 Set $\omega_{u,\ell}^{k^*}(t) = 1$ and $\omega_{u,\ell}^k(t) = 0, \forall k \neq k^*$;
- 14 **end**
- 15 **end**
- 16 Execute action a_t , observe reward r_t and next state s_{t+1} ;
- 17 Store transition (s_t, a_t, r_t, s_{t+1}) into $\mathcal{D}$;
- 18 Sample a minibatch of N_b transitions from $\mathcal{D}$;
- 19 Compute target:
- 20 $y_i = r_i + \gamma Q'(s_{i+1}, \mu'(s_{i+1}|\theta^{\mu'})|\theta^{Q'})$;
- 21 Update the critic by minimizing $L(\theta^Q) = \frac{1}{N_b} \sum_i (y_i - Q(s_i, a_i|\theta^Q))^2$;
- 22 Update actor using policy gradient;
- 23 Soft update targets: $\theta^{Q'} \leftarrow \psi \theta^Q + (1 - \psi) \theta^{Q'}$, $\theta^{\mu'} \leftarrow \psi \theta^\mu + (1 - \psi) \theta^{\mu'}$;
- 24 **end**

UAV baseline, while the proposed battery-aware weighted Voronoi scheme is utilized for the dual-UAV system.

- **Step 2: Online DRL Training (P2).** Based on the fixed UAV trajectories and service partitions from P1, the DRL agent is trained to jointly optimize the dual-domain SWIPT variables (i.e., time-switching factor τ , transmit power $\mathbf{p}$, scheduling ω , RIS phase shifts Θ).
- **Step 3: Testing and Statistical Evaluation.** To evaluate the generalization capability, 100 independent sets of random UT mobility trajectories are generated, which strictly differ from the routes used during training. The pre-trained DRL model from Step 2 is then executed on these unseen testing routes, and the final performance

TABLE II: Simulation Parameters

Symbol	Description	Value
<i>Environment & Channel Parameters</i>		
$\mathcal{A}, \mathcal{B}$	Environment constants of LoS	9.61, 0.16
α	Path-loss exponent (AP-RIS)	3.0
$\bar{\alpha}$	Path-loss exponent (RIS-UT)	2.5
κ	Reference path-loss at 1 m	-30 dB
κ_{NLoS}	NLoS attenuation factor	20 dB
σ_k^2	Noise power	-102 dBm
<i>System Configurations</i>		
H_u	UAV operational altitude	20 m
Z	Number of AP antennas	8
L	RIS elements per UAV	16
P_{max}	Max. AP transmit power	500 W
η	Energy harvesting efficiency	0.7
Γ_{min}	Min. QoS throughput requirement	70 Mbps

TABLE III: Performance Comparison of Different DRL Algorithms under Single-UAV and Dual-UAV Deployments

Algorithm	EH Eff. (%)		Sum Tput (Mbps)		Min. Tput (Mbps)	
	Single	Dual	Single	Dual	Single	Dual
Proposed DDPG	50.31	54.06	598.18	803.01	44.44	86.38
TD3	44.53	51.56	616.10	819.63	47.77	94.81
SAC	45.03	49.62	676.26	809.80	73.25	88.04
PPO	35.53	40.41	630.22	884.19	0.00	109.10

metrics are obtained by averaging the statistical results.

Unless otherwise specified, the UAVs operate at a fixed altitude, and the UTs move within a square area of $40 \text{ m} \times 40 \text{ m}$. The detailed simulation parameters are listed in Table II. Specifically, the channel and environment configurations are set according to standard UAV and RIS literature [9], [10], while the minimum QoS throughput requirement (Γ_{min}) is adopted from [10].

A. UAV Trajectory Illustration

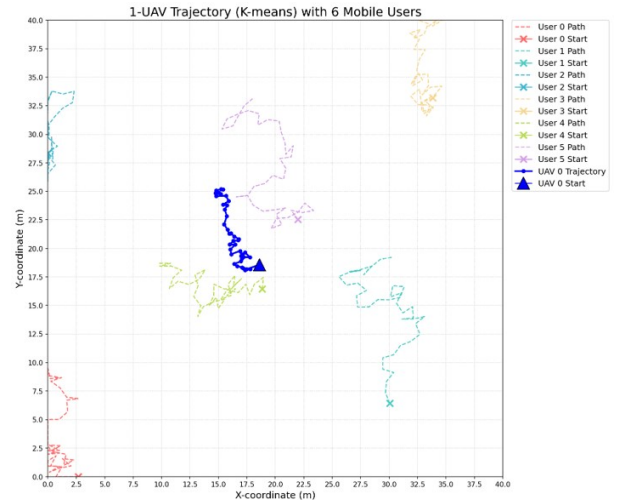


Fig. 6: UAV trajectory generated by the density-aware (K-means) deployment for the single-UAV case with mobile UTs.

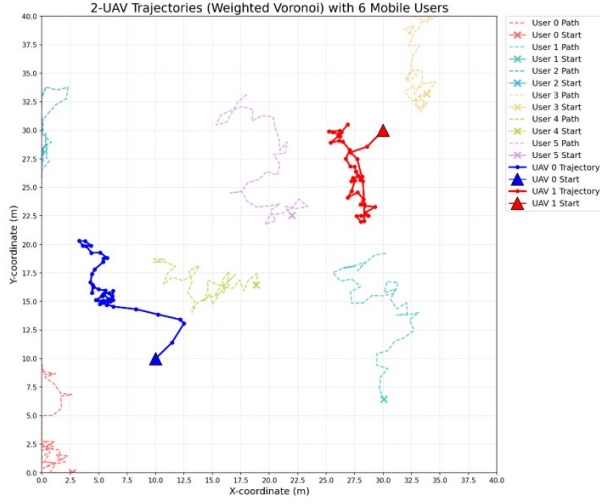


Fig. 7: Trajectories of dual UAVs generated by the proposed battery-aware weighted Voronoi deployment.

Fig. 6 illustrates the UAV trajectory generated by the density-aware deployment for the single-UAV case under mobile UTs. The UAV dynamically adjusts its position to follow the spatial distribution of users, serving as the training trajectory for the online DRL phase.

Fig. 7 depicts the trajectories of the two UAVs obtained by the proposed battery-aware weighted Voronoi deployment. Compared with the single-UAV case, the dual-UAV system exhibits clear spatial separation between the service regions of the two UAVs, effectively balancing user load and reducing overlapping coverage.

B. Energy Harvesting Efficiency Performance

Fig. 3 shows the energy harvesting (EH) efficiency per time slot for the single-UAV and dual-UAV systems under different DRL algorithms. The dual-UAV deployment consistently achieves higher EH efficiency than the single-UAV baseline due to reduced propagation loss and improved spatial resource utilization.

Among all considered DRL methods, **DDPG achieves the highest EH efficiency** in both deployment scenarios. Specifically, the average EH efficiency of DDPG increases from **50.31%** in the single-UAV case to **54.06%** in the dual-UAV case, corresponding to a relative improvement of approximately **7.5%**. This performance gain highlights the effectiveness of combining the proposed battery-aware weighted Voronoi deployment with DDPG's deterministic policy learning.

In contrast, PPO [35] exhibits the lowest EH efficiency due to its on-policy nature, while SAC [36] and TD3 [37] achieve moderate performance improvements under the dual-UAV deployment. These results indicate that DDPG is particularly well-suited for EH-oriented SWIPT optimization in UAV-assisted networks.

C. System Throughput Performance

Fig. 4 presents the system sum throughput per time slot for both single-UAV and dual-UAV systems. Although the

primary objective of the proposed framework is to maximize EH efficiency, the dual-UAV system also achieves significantly higher throughput compared with the single-UAV case.

For the DDPG-based scheme, the average system throughput increases from **598.18 Mbps** in the single-UAV case to **803.01 Mbps** in the dual-UAV case, representing a gain of approximately **34.2%**. This improvement is mainly attributed to the reduced average UAV-UT distance and the more balanced user association enabled by the weighted Voronoi partitioning.

Although PPO achieves a slightly higher sum throughput in the dual-UAV scenario (**884.19 Mbps**), this comes at the expense of severely degraded EH efficiency and, critically, complete service starvation for the worst-served user in the single-UAV case, as discussed in the following subsection.

D. Worst-User Throughput and Fairness

To further evaluate service fairness, Fig. 5 shows the worst-user throughput under different deployment strategies and algorithms. The dual-UAV system significantly improves the minimum achievable throughput across users, indicating enhanced fairness.

A particularly striking observation is that **PPO achieves a worst-user throughput of 0.00 Mbps** in the single-UAV scenario, meaning at least one user receives no service whatsoever. In contrast, the proposed DDPG-based scheme sustains a non-zero worst-user throughput of **44.44 Mbps** under the same deployment, demonstrating its inherent robustness against user starvation. Similarly, while TD3 achieves a higher worst-user throughput in the dual-UAV scenario (**94.81 Mbps** versus **86.38 Mbps** for DDPG), this marginal gain is accompanied by a noticeable reduction in EH efficiency (from 54.06% to 51.56%), which conflicts with the primary optimization objective.

Overall, the proposed DDPG-based scheme maintains a consistently non-zero worst-user throughput across *both* single-UAV and dual-UAV deployments while simultaneously achieving the highest EH efficiency. This demonstrates that DDPG provides a more balanced and deployment-robust trade-off among energy efficiency, throughput, and fairness than all other evaluated algorithms.

E. Training Convergence Behavior

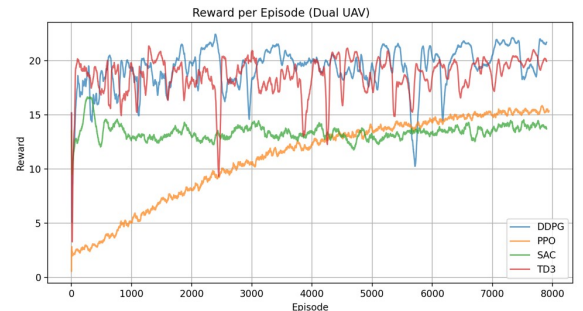


Fig. 8: Reward per training episode for different DRL algorithms under the dual-UAV deployment.

Fig. 8 illustrates the convergence behavior of the proposed DDPG algorithm compared with other baseline DRL methods. Several critical observations can be drawn from the reward curves. First and most importantly, the proposed **DDPG agent achieves the highest final reward** and exhibits stable convergence after sufficient training episodes. This confirms its superior capability to efficiently discover the optimal resource allocation policy in the highly complex, continuous-action space of the UAV-assisted SWIPT system.

Second, while PPO displays a steady upward learning trend, its convergence speed is significantly slower due to its on-policy nature, requiring vastly more environmental interactions to update its policy. Third, although SAC and TD3 show rapid initial improvements, their reward curves eventually plateau at substantially lower levels, indicating that they may become trapped in suboptimal policies. Overall, the training curves clearly demonstrate that DDPG has the best learning efficiency and the highest steady-state performance among the evaluated algorithms.

F. Performance Summary

Table III summarizes the average EH efficiency, system throughput, and worst-user throughput for all evaluated algorithms under single-UAV and dual-UAV deployments.

Overall, the proposed DDPG-based dual-UAV framework achieves the best trade-off among energy efficiency, throughput, and fairness. Compared with the single-UAV baseline, the dual-UAV DDPG scheme improves EH efficiency from **50.31%** to **54.06%** (+7.5%), increases system throughput by approximately **34.2%** (from 598.18 Mbps to 803.01 Mbps), and nearly doubles the worst-user throughput (from 44.44 Mbps to 86.38 Mbps). Crucially, the proposed scheme is the only algorithm that guarantees non-zero worst-user throughput in the challenging single-UAV scenario, which none of the competing baselines can achieve simultaneously with high EH efficiency. These results clearly validate the effectiveness of the proposed battery-aware weighted Voronoi deployment combined with DRL-based SWIPT optimization.

VIII. CONCLUSION AND FUTURE WORK

In this work, the critical challenges of limited battery endurance and load imbalance in multi-UAV RIS-assisted networks were investigated. To address these issues, a novel two-stage framework was proposed, coupling offline geometry design with online signal-level optimization. In the first stage, a battery-aware weighted Voronoi partitioning strategy was developed to dynamically adjust service regions based on user distribution and residual energy, thereby achieving proactive load balancing. In the second stage, a DDPG-based robust resource allocation scheme was designed to optimize the dual-domain SWIPT parameters, including time-switching factors, transmit power, and RIS element scheduling, under the constraints of the offline partitions.

Simulation results demonstrated the superiority of the proposed dual-UAV architecture over conventional single-UAV benchmarks. The integration of weighted Voronoi partitioning with DDPG-based control not only significantly enhanced

the energy harvesting efficiency but also improved system throughput and worst-case user fairness. Furthermore, the proposed framework was shown to effectively mitigate the energy exhaustion problem common in heuristic trajectory designs by allowing energy-abundant UAVs to relieve the burden on energy-deficient ones.

In future work, we plan to extend this centralized training framework to a decentralized multi-agent deep reinforcement learning (MADRL) architecture to support larger-scale UAV swarms. Additionally, investigating 3D spatial placement and trajectory optimization, as well as considering imperfect channel state information (CSI) in the weighted Voronoi tessellation, would be valuable directions. Finally, incorporating non-linear energy harvesting circuit models and practical hardware impairments—such as discrete RIS phase shifts—will be critical next steps to further enhance the practical deployment viability of the proposed framework.

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